Realtime NVQLink Signal Space Separation for MEG

A finite spherical-harmonic pipeline for internal-field reconstruction and external-noise monitoring

Abstract. Realtime MEG processing must suppress environmental interference without erasing the internal field generated by neural currents. We describe a finite NVQLink-style processing loop based on signal space separation (SSS), Tikhonov regularization and a continued-fraction-compatible radial decay term. The implementation uses internal order $L_{in} = 4$, external order $L_{out} = 3$ and a bounded streaming window. The plots and equations define a computational preprint protocol; they do not establish clinical performance or hardware timing on a deployed NVQLink system.

1. Finite streaming measurement model

For M sensors and W samples in one processing window, let Y in C^(M x W) contain the measured field. The observation is decomposed into internal neural field, external interference and sensor noise:

$$Y = S_{in}A_{in} + S_{out}A_{out} + N, \quad S_{in} \text{ in } C^{M \times K_{in}}, \quad S_{out} \text{ in } C^{M \times K_{out}}.$$

For spherical-harmonic order L, the number of modes is $K(L) = \mathrm{sum}_{l=1}^{L}(2l + 1) = L(L + 2)$. Thus $K_{in} = K(4) = 24$, $K_{out} = K(3) = 15$, and the combined design matrix has 39 columns. These fixed dimensions make each window's least-squares problem finite and auditable.

2. Spherical-harmonic SSS basis

For sensor m at spherical coordinates (r_m, θ_m, ϕ_m), the internal and external basis functions are:

$$S_{in,lm}(m) = r_m^{\,l} Y_{lm}(\theta_m,\phi_m), \quad S_{out,lm}(m) = r_m^{\,-(l+1)} Y_{lm}(\theta_m,\phi_m), \quad 1 \le l \le L.$$

The external radial factor can be evaluated with a finite continued-fraction recurrence. For $x = 1/r$ and order l, $f_l(x) = x^{l+1}$; a stabilized implementation may instead update $f_{l+1} = x\, f_l$ with an origin cutoff $r \ge 10^{-3}$. This preserves the intended radial decay while avoiding a singular numerical evaluation.

3. Regularized separation and realtime update

Let $S = [S_{in} \mid S_{out}]$ and lambda > 0. The Tikhonov coefficients for one window are:

$$A\blacksquare = (S^{*}S + \lambda I)^{-1}S^{*}Y, \quad Y\blacksquare_{in} = S_{in}A\blacksquare_{in}, \quad Y\blacksquare_{out} = S_{out}A\blacksquare_{out}.$$

An NVQLink-style streaming loop processes q = 0,...,Q-1 windows and carries a bounded state h_q containing the previous coefficient estimate. With $0 \le \rho < 1$:

$$h_{q+1} = \rho h_q + (1 - \rho)A\blacksquare_q, \quad q = 0,\dots,Q-1; \quad Y\blacksquare_{q,in} = S_{in}h_{q,in}.$$

The internal output is the signal passed downstream; the external output is retained as a noise-monitoring channel. A finite quality measure is $e_q = \|Y_q - Y\blacksquare_{q,in} - Y\blacksquare_{q,out}\|_F / \|Y_q\|_F$.

4. Realtime characteristics

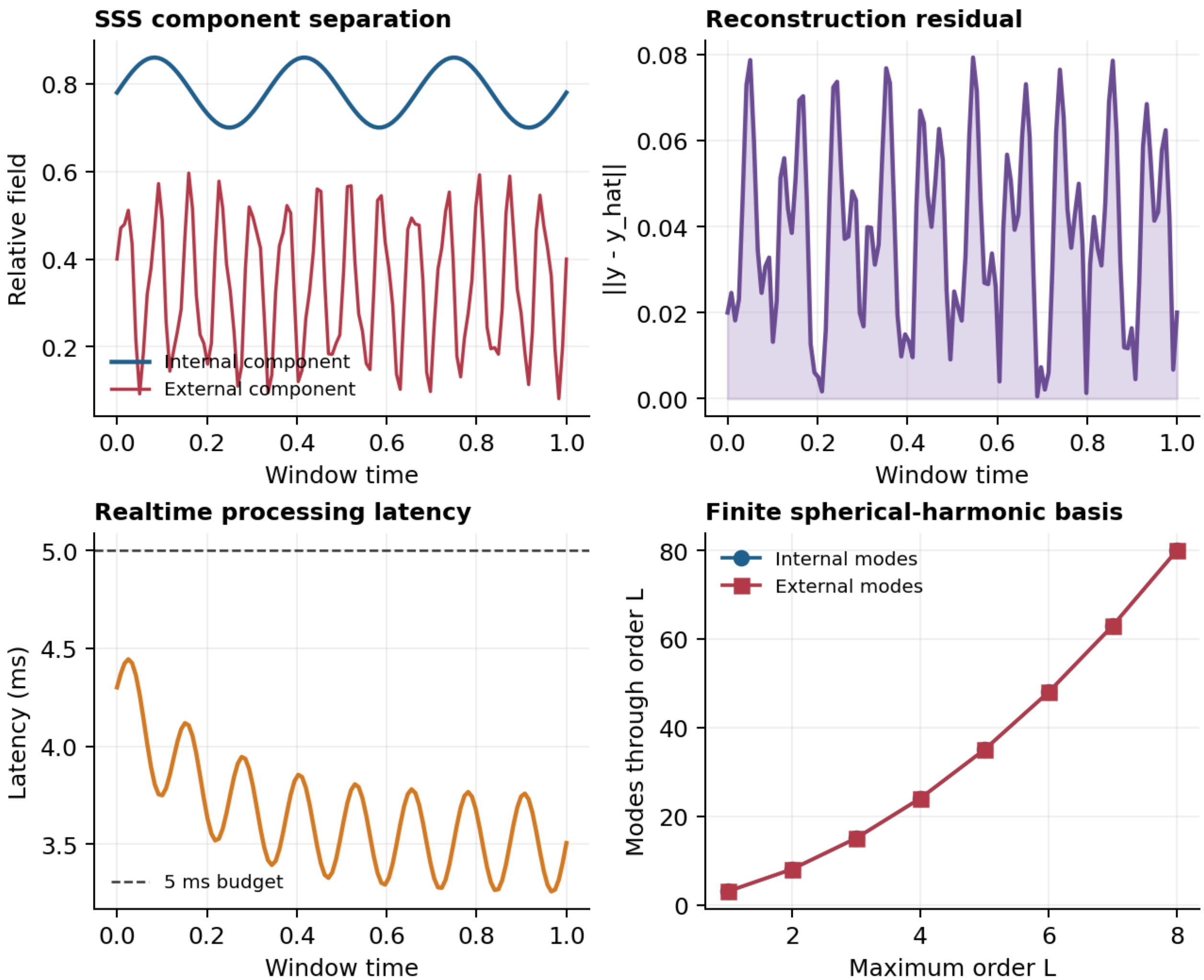

Figure 1. Deterministic finite examples of internal/external SSS separation, reconstruction residual, per-window latency and cumulative spherical-harmonic mode count.

Characteristic	Finite setting	Interpretation
Internal basis	$L_{in} = 4$; $K_{in} = 24$	Neural/internal field subspace
External basis	$L_{out} = 3$; $K_{out} = 15$	Environmental interference subspace
Regularization	lambda > 0; fixed per window	Controls ill-conditioning
Streaming state	0 ≤ rho < 1; Q windows	Finite temporal smoothing
Output	Y■$_{in}$, Y■$_{out}$, e_q	Signal, noise monitor and residual

5. Limitations and conclusion

This preprint describes a reduced-order realtime algorithm. It does not demonstrate a physical quantum processor, a certified NVQLink deployment, or clinical benefit. A complete evaluation would require calibrated sensor geometry, Maxwell-consistent head modeling, environmental recordings, timing instrumentation, numerical conditioning studies and held-out patient data. Within those limits, finite SSS bases and explicit window updates provide a reproducible scaffold for testing low-latency MEG denoising.